SECTION 5

Subtracting Decimals

Line up the decimal points, subtract each place, and regroup when the top digit is too small.

SKILL One new idea at a time.

SHOW HOW Watch a problem solved step by step.

TRY ONE Finish a problem with help.

YOUR TURN Solve problems on your own.

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Name _____ Started _____

Georgia K-12 Mathematics Standards 5.NR.4.4 (subtract decimals to the hundredths place using a variety of strategies).

5.1 Line Up and Subtract

Goal: I can subtract decimals place by place.

SHOW HOW

Subtract 6.85 − 2.43.

```
  6.85
− 2.43
------
  4.42
```

1. **Line up the decimal points.** Bigger number on top.
2. Start at the **right**. Hundredths: 5 − 3 = **2**.
3. Tenths: 8 − 4 = **4**. Ones: 6 − 2 = **4**.
4. Bring the decimal point straight down: **4.42**

Why it works: same as adding: lining up the points means you take hundredths from hundredths and tenths from tenths.

LET'S TRY ONE

Subtract 7.68 − 3.25. Fill in the boxes.

```
  7.68
− 3.25
------
  [ ].[ ][ ]
```

Hundredths: _____ Tenths: _____ Ones: _____

Difference: _____

YOUR TURN — SKILL 5.1 · SUBTRACT. FILL IN THE BOXES.

1

$$\begin{array}{r} 0.8 \\ -\ 0.5 \\ \hline \square.\square \end{array}$$

2

$$\begin{array}{r} 0.76 \\ -\ 0.34 \\ \hline \square.\square\square \end{array}$$

3

$$\begin{array}{r} 5.68 \\ -\ 2.15 \\ \hline \square.\square\square \end{array}$$

4

$$\begin{array}{r} 9.47 \\ -\ 6.23 \\ \hline \square.\square\square \end{array}$$

5

$$\begin{array}{r} 14.9 \\ -\ 3.6 \\ \hline \square\square.\square \end{array}$$

6

$$\begin{array}{r} 0.87 \\ -\ 0.43 \\ \hline \square.\square\square \end{array}$$

7 Write it in columns, then subtract:
8.79 − 4.56

Difference _____

8 Write it in columns, then subtract:
25.65 − 13.42

Difference _____

5.2 Subtracting with Regrouping

Goal: I can regroup 1 of a place as 10 of the next smaller place.

SHOW HOW

Solve 8.20 − 3.76.

7 11 10
~~8~~ . ~~2~~ ~~0~~
− 3 . 7 6
4 . 4 4

Red numbers show each regroup. The tenths went 2 → 1 (gave 1 tenth to the hundredths) → 11 (got 10 tenths from a one).

1. **Line up the points.** Start with the **hundredths**: 0 − 6? Not enough.
2. **Regroup a tenth:** 2 tenths become **1** tenth; the hundredths get 10. Now 10 − 6 = **4**.
3. **Tenths**: 1 − 7? Not enough. **Regroup a one:** 8 ones become **7**; the tenths get 10 more: 11 − 7 = **4**.
4. **Ones**: 7 − 3 = **4**. Bring the point down: **4.44**

Why it works: 1 one = 10 tenths and 1 tenth = 10 hundredths. Regrouping trades one bigger piece for ten smaller pieces, so the top number keeps the same value: 8.20 = 7 ones + 11 tenths + 10 hundredths.

LET'S TRY ONE

Solve 5.42 − 2.17. Cross out and regroup above the digits.

□ □ □
5 . 4 2
− 2 . 1 7
□ . □ □

Hundredths: 2 − 7? Not enough, so regroup a tenth.

4 tenths → _____ tenths

2 hundredths → _____ hundredths

_____ − 7 = _____ Difference: _____

YOUR TURN — SKILL 5.2 · SUBTRACT. REGROUP WHEN YOU NEED TO.

1

$$\begin{array}{r} 0.72 \\ -\ 0.38 \\ \hline \square.\square\square \end{array}$$

2

$$\begin{array}{r} 4.53 \\ -\ 1.29 \\ \hline \square.\square\square \end{array}$$

3

$$\begin{array}{r} 6.4 \\ -\ 2.7 \\ \hline \square.\square \end{array}$$

4

$$\begin{array}{r} 7.25 \\ -\ 3.68 \\ \hline \square.\square\square \end{array}$$

5 Write it in columns, then subtract:
9.13 − 4.57

Difference _____

6 Write it in columns, then subtract:
12.3 − 5.8

Difference _____

7 Write it in columns, then subtract:
8.31 − 2.75

Difference _____

8 Write it in columns, then subtract:
11.42 − 6.87

Difference _____

5.3 Placeholder Zeros and Different Lengths

Goal: I can subtract when the numbers have different numbers of decimal places.

SHOW HOW

Subtract 5 − 2.36.

```
   4   9  10
   5 . 0  0
 − 2 . 3  6
 ---------
   2 . 6  4
```

1. A whole number has its point at the end: **5 = 5.00**. Write the zeros so both numbers have hundredths.
2. Hundredths: 0 − 6? The tenths are 0 too, so **regroup a one**: 5 ones → 4 ones and 10 tenths.
3. Now regroup a tenth: 10 tenths → **9 tenths** and 10 hundredths.
4. Subtract: 10 − 6 = 4, 9 − 3 = 6, 4 − 2 = 2 → **2.64**

Why it works: 5.00 has the same value as 5 (Section 1). The zeros give you places to regroup into. Never take the bottom number from the top just because it is "bigger-looking".

LET'S TRY ONE

Subtract 4.5 − 1.27. Write the placeholder zero first.

```
   □  □  □
   4 . 5
 − 1 . 2  7
 ---------
   □ . □  □
```

4.5 = _____

Difference: _____

YOUR TURN — SKILL 5.3 · WRITE ZEROS, THEN SUBTRACT

1 3.6 − 1.25

Difference _____

2 0.9 − 0.45

Difference _____

3 8 − 3.4

Difference _____

4 7.25 − 2.8

Difference _____

5 6 − 2.75

Difference _____

6 1 − 0.36

Difference _____

7 10 − 4.05

Difference _____

8 4.2 − 1.87

Difference _____

5.4 Check Subtraction by Adding

Goal: I can use addition to check a subtraction answer.

SHOW HOW

Check: is 8.20 − 3.76 = 4.44 correct?

```
  1   1
  4.44
+ 3.76
------
  8.20
```

1. Subtraction takes a part away from the whole: **8.20 − 3.76 = 4.44**.
2. So the two parts must add back to the whole: **4.44 + 3.76** should equal 8.20.
3. Add: 4.44 + 3.76 = **8.20**. It matches, so the subtraction is **correct**.

Why it works: answer + number you subtracted = number you started with. **If it does not match, fix it:** 5 − 1.2 = 4.2? Check 4.2 + 1.2 = 5.4, not 5. Subtract again: 5.0 − 1.2 = 3.8, and 3.8 + 1.2 = 5 ✓.

LET'S TRY ONE

Check 6.23 − 2.75 = 3.48 by adding.

```
  □  □  □
  3.48
+ 2.75
------
  □.□□
```

3.48 + 2.75 = _____

Does it match 6.23? _____

YOUR TURN — SKILL 5.4 · SUBTRACT, THEN CHECK BY ADDING

1 **5.83 − 2.41**

Check: answer + 2.41

Difference _____

2 **7.3 − 4.6**

Check: answer + 4.6

Difference _____

3 **9.15 − 3.48**

Check: answer + 3.48

Difference _____

4 **6 − 1.72**

Check: answer + 1.72

Difference _____

5 Is **4.49 − 1.95 = 2.54** correct? Check by adding.

Correct or not? _____

6 Is **6.2 − 2.87 = 3.43** correct? Check by adding. If not, fix it.

Correct? Fixed answer: _____

7 Is **8 − 3.35 = 4.65** correct? Check by adding.

Correct or not? _____

8 Is **5 − 3.12 = 2.12** correct? Check by adding. If not, fix it.

Correct? Fixed answer: _____

5.5 Subtraction Word Problems

Goal: I can tell when a story needs subtracting, then subtract and label.

SHOW HOW

Mia had $10.00. She spent $6.45. How much money does she have left?

1. **What is asked?** How much is **left**. "Left", "how much more", and "difference" mean **subtract**.
2. **Estimate:** 10 − 6 = about **$4**.
3. **Line up and subtract:** 10.00 − 6.45 = **3.55** (regroup across the zeros).
4. **Check and label:** 3.55 + 6.45 = 10.00 ✓. Mia has **$3.55** left.

```
  0  9   9  10
  1  0 . 0  0
−    6 . 4  5
-------------
     3 . 5  5
```

Subtracting words: left, remain, how much more, how much less, how much taller/longer/heavier, difference, change.

LET'S TRY ONE

Dad is 1.62 m tall. Gabriella is 1.35 m tall. How much taller is Dad?

Operation: _____ Exact: _____

Check by adding: _____ Answer with units: _____

YOUR TURN

SKILL 5.5 · SOLVE. LABEL YOUR ANSWER.

1 Leo pays for a **$2.75** drink with a **$5** bill. How much change does he get?

$ _____

2 A bottle has **2.4 L** of water. Ava drinks **0.85 L**. How much water is left?

Answer _____ L

3 A pencil is **12.6 cm** long. A crayon is **9.35 cm** long. How much longer is the pencil?

Answer _____ cm

4 A puppy weighed **2.85 lb**. Now it weighs **4.5 lb**. How much weight did it gain?

Answer _____ lb

5 Sam ran in **10.2 seconds**. Jo ran in **9.87 seconds**. What is the difference in their times?

Answer _____ seconds

6 Mom has **$20**. Groceries cost **$13.48**. How much money is left?

$ _____

7 A bag of rice weighs **3.12 kg**. Mom uses **1.6 kg**. How much rice is left?

Answer _____ kg

8 A race is **8.4 miles**. Kai has run **5.75 miles**. How far does he still have to go?

Answer _____ mi

Section 5 Review

Mixed practice from Skills 5.1 to 5.5. No hints: show what you know.

YOUR TURN **REVIEW · PROBLEMS 1–8**

1 **0.96 − 0.42**

Difference _____

2 **7.34 − 2.58**

Difference _____

3 **9.5 − 3.75**

Difference _____

4 **12 − 6.08**

Difference _____

5 **5.06 − 2.47**

Difference _____

6 Is **6.32 − 3.64 = 3.78** correct? Check by adding.

Correct or not? _____

7 A kite costs **$8.65**. Gabriella pays with **$15**. How much change does she get?

$ _____

8 A board is **2.3 m** long. Dad cuts off **1.85 m**. How long is the piece that is left?

Answer _____ m

Parent Answer Key

Equal values are correct (3.40 = 3.4). The steps show each regroup. “Try” is the Let's Try One problem.

5.1 Line Up and Subtract (No Regrouping)

Try **4.43** Line up the points. Hundredths: 8 − 5 = 3. Tenths: 6 − 2 = 4. Ones: 7 − 3 = 4. → 4.43.

1 **0.3** Line up the points. Tenths: 8 − 5 = 3. → 0.3.

2 **0.42** Line up the points. Hundredths: 6 − 4 = 2. Tenths: 7 − 3 = 4. → 0.42.

3 **3.53** Line up the points. Hundredths: 8 − 5 = 3. Tenths: 6 − 1 = 5. Ones: 5 − 2 = 3. → 3.53.

4 **3.24** Line up the points. Hundredths: 7 − 3 = 4. Tenths: 4 − 2 = 2. Ones: 9 − 6 = 3. → 3.24.

5 **11.3** Line up the points. Tenths: 9 − 6 = 3. Ones: 4 − 3 = 1. Tens: 1 − 0 = 1. → 11.3.

6 **0.44** Line up the points. Hundredths: 7 − 3 = 4. Tenths: 8 − 4 = 4. → 0.44.

7 **4.23** Line up the points. Hundredths: 9 − 6 = 3. Tenths: 7 − 5 = 2. Ones: 8 − 4 = 4. → 4.23.

8 **12.23** Line up the points. Hundredths: 5 − 2 = 3. Tenths: 6 − 4 = 2. Ones: 5 − 3 = 2. Tens: 2 − 1 = 1. → 12.23.

5.2 Subtracting with Regrouping

Try **3.25** Line up the points. Hundredths: 2 < 7, regroup: 12 − 7 = 5. Tenths (4 became 3): 3 − 1 = 2. Ones: 5 − 2 = 3. → 3.25.

1 **0.34** Line up the points. Hundredths: 2 < 8, regroup: 12 − 8 = 4. Tenths (7 became 6): 6 − 3 = 3. → 0.34.

2 **3.24** Line up the points. Hundredths: 3 < 9, regroup: 13 − 9 = 4. Tenths (5 became 4): 4 − 2 = 2. Ones: 4 − 1 = 3. → 3.24.

3 **3.7** Line up the points. Tenths: 4 < 7, regroup: 14 − 7 = 7. Ones (6 became 5): 5 − 2 = 3. → 3.7.

4 **3.57** Line up the points. Hundredths: 5 < 8, regroup: 15 − 8 = 7. Tenths (2 became 1): 1 < 6, regroup: 11 − 6 = 5. Ones (7 became 6): 6 − 3 = 3. → 3.57.

5 **4.56** Line up the points. Hundredths: 3 < 7, regroup: 13 − 7 = 6. Tenths (1 became 0): 0 < 5, regroup: 10 − 5 = 5. Ones (9 became 8): 8 − 4 = 4. → 4.56.

6 **6.5** Line up the points. Tenths: 3 < 8, regroup: 13 − 8 = 5. Ones (2 became 1): 1 < 5, regroup: 11 − 5 = 6. → 6.5.

7 **5.56** Line up the points. Hundredths: 1 < 5, regroup: 11 − 5 = 6. Tenths (3 became 2): 2 < 7, regroup: 12 − 7 = 5. Ones (8 became 7): 7 − 2 = 5. → 5.56.

8 **4.55** Line up the points. Hundredths: 2 < 7, regroup: 12 − 7 = 5. Tenths (4 became 3): 3 < 8, regroup: 13 − 8 = 5. Ones (1 became 0): 0 < 6, regroup: 10 − 6 = 4. → 4.55.

5.3 Placeholder Zeros and Different Lengths

Try **3.23** Line up the points; write 4.50 − 1.27. Hundredths: 0 < 7, regroup: 10 − 7 = 3. Tenths (5 became 4): 4 − 2 = 2. Ones: 4 − 1 = 3. → 3.23.

1 **2.35** Line up the points; write 3.60 − 1.25. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (6 became 5): 5 − 2 = 3. Ones: 3 − 1 = 2. → 2.35.

2 **0.45** Line up the points; write 0.90 − 0.45. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (9 became 8): 8 − 4 = 4. → 0.45.

3 **4.6** Line up the points; write 8.0 − 3.4. Tenths: 0 < 4, regroup: 10 − 4 = 6. Ones (8 became 7): 7 − 3 = 4. → 4.6.

4 **4.45** Line up the points; write 7.25 − 2.80. Hundredths: 5 − 0 = 5. Tenths: 2 < 8, regroup: 12 − 8 = 4. Ones (7 became 6): 6 − 2 = 4. → 4.45.

5 **3.25** Line up the points; write 6.00 − 2.75. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 7 = 2. Ones (6 became 5): 5 − 2 = 3. → 3.25.

6 **0.64** Line up the points; write 1.00 − 0.36. Hundredths: 0 < 6, regroup: 10 − 6 = 4. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 3 = 6. → 0.64.

7 **5.95** Line up the points; write 10.00 − 4.05. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 0 = 9. Ones: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 4 = 5. → 5.95.

8 **2.33** Line up the points; write 4.20 − 1.87. Hundredths: 0 < 7, regroup: 10 − 7 = 3. Tenths (2 became 1): 1 < 8, regroup: 11 − 8 = 3. Ones (4 became 3): 3 − 1 = 2. → 2.33.

5.4 Check Subtraction by Adding

Try **6.23** 3.48 + 2.75 = 6.23, which matches 6.23, so 6.23 − 2.75 = 3.48 is correct.

1 **3.42** Line up the points. Hundredths: 3 − 1 = 2. Tenths: 8 − 4 = 4. Ones: 5 − 2 = 3. → 3.42.

2 **2.7** Line up the points. Tenths: 3 < 6, regroup: 13 − 6 = 7. Ones (7 became 6): 6 − 4 = 2. → 2.7.

3 **5.67** Line up the points. Hundredths: 5 < 8, regroup: 15 − 8 = 7. Tenths (1 became 0): 0 < 4, regroup: 10 − 4 = 6. Ones (9 became 8): 8 − 3 = 5. → 5.67.

4 **4.28** Line up the points; write 6.00 − 1.72. Hundredths: 0 < 2, regroup: 10 − 2 = 8. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 7 = 2. Ones (6 became 5): 5 − 1 = 4. → 4.28.

5 **Correct** 2.54 + 1.95 = 4.49, which matches 4.49.

6 **Not correct (3.33)** 3.43 + 2.87 = 6.30, not 6.2. The correct answer: 6.20 − 2.87 = 3.33 (check: 3.33 + 2.87 = 6.20).

7 **Correct** 4.65 + 3.35 = 8.00 = 8.

8 **Not correct (1.88)** 2.12 + 3.12 = 5.24, not 5. Correct: 5.00 − 3.12 = 1.88 (check: 1.88 + 3.12 = 5.00). The student subtracted the smaller digit from the bigger one in each column.

5.5 Subtraction Word Problems

Try **0.27** How much taller means subtract. 1.62 − 1.35 = 0.27 m. Check: 0.27 + 1.35 = 1.62.

1 **$2.25** Line up the points; write 5.00 − 2.75. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 7 = 2. Ones (5 became 4): 4 − 2 = 2. → 2.25.

2 **1.55 L** Line up the points; write 2.40 − 0.85. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (4 became 3): 3 < 8, regroup: 13 − 8 = 5. Ones (2 became 1): 1 − 0 = 1. → 1.55.

3 **3.25 cm** Line up the points; write 12.60 − 9.35. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (6 became 5): 5 − 3 = 2. Ones: 2 < 9, regroup: 12 − 9 = 3. → 3.25.

4 **1.65 lb** Line up the points; write 4.50 − 2.85. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (5 became 4): 4 < 8, regroup: 14 − 8 = 6. Ones (4 became 3): 3 − 2 = 1. → 1.65.

5 **0.33 seconds** Line up the points; write 10.20 − 9.87. Hundredths: 0 < 7, regroup: 10 − 7 = 3. Tenths (2 became 1): 1 < 8, regroup: 11 − 8 = 3. Ones: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 9 = 0. → 0.33.

6 **$6.52** Line up the points; write 20.00 − 13.48. Hundredths: 0 < 8, regroup: 10 − 8 = 2. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 4 = 5. Ones: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 3 = 6. Tens (2 became 1): 1 − 1 = 0. → 6.52.

7 **1.52 kg** Line up the points; write 3.12 − 1.60. Hundredths: 2 − 0 = 2. Tenths: 1 < 6, regroup: 11 − 6 = 5. Ones (3 became 2): 2 − 1 = 1. → 1.52.

8 **2.65 mi** Line up the points; write 8.40 − 5.75. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (4 became 3): 3 < 7, regroup: 13 − 7 = 6. Ones (8 became 7): 7 − 5 = 2. → 2.65.

Review Section 5 Review

1 **0.54** Line up the points. Hundredths: 6 − 2 = 4. Tenths: 9 − 4 = 5. → 0.54.

2 **4.76** Line up the points. Hundredths: 4 < 8, regroup: 14 − 8 = 6. Tenths (3 became 2): 2 < 5, regroup: 12 − 5 = 7. Ones (7 became 6): 6 − 2 = 4. → 4.76.

3 **5.75** Line up the points; write 9.50 − 3.75. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (5 became 4): 4 < 7, regroup: 14 − 7 = 7. Ones (9 became 8): 8 − 3 = 5. → 5.75.

4 **5.92** Line up the points; write 12.00 − 6.08. Hundredths: 0 < 8, regroup: 10 − 8 = 2. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 0 = 9. Ones (2 became 1): 1 < 6, regroup: 11 − 6 = 5. → 5.92.

5 **2.59** Line up the points. Hundredths: 6 < 7, regroup: 16 − 7 = 9. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 4 = 5. Ones (5 became 4): 4 − 2 = 2. → 2.59.

6 **Not correct (2.68)** 3.78 + 3.64 = 7.42, not 6.32. Correct: 6.32 − 3.64 = 2.68 (check: 2.68 + 3.64 = 6.32).

7 **$6.35** Line up the points; write 15.00 − 8.65. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 6 = 3. Ones (5 became 4): 4 < 8, regroup: 14 − 8 = 6. → 6.35.

8 **0.45 m** Line up the points; write 2.30 − 1.85. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (3 became 2): 2 < 8, regroup: 12 − 8 = 4. Ones (2 became 1): 1 − 1 = 0. → 0.45.